



Figure 1: Rajnikanth's robot

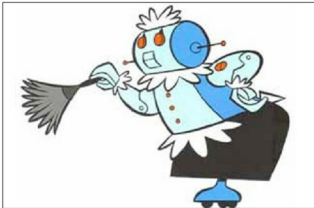


Figure 3: Rosy, the robot maid

Microsoft led to the Microsoft Robotics Developer Studio, which also allowed simulation in three dimensions, and enabled users to design their own robots. However, being a Microsoft product, it never had the opportunity and the reach that open source has.

The route to Robot Operating System (ROS)

ROS was developed by Willow Garage, a consortium of American Universities, in 2007. The motivation was



Figure 2: Wall-E

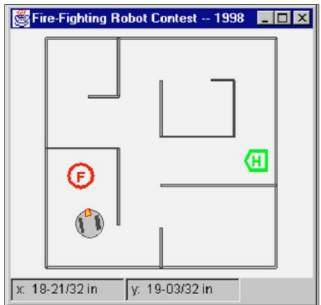


Figure 4: Rossum's Playhouse, an early robot simulator

to develop a meta interface for robotics software. ROS embraced open source, and initially forked applications from Player Project and other open source robotic applications. It was stable, easier to install, had good documentation, an easy-to-understand user interface, and behaved more like an operating system. In the five years of its existence, ROS has had four releases, and has incorporated many interesting open source graphical simulation interfaces and control paradigms. ROS has come to be the tool for many researchers and academics, who have contributed their code to the ROS community. At the time of writing, other than the ROS website, <http://www.ros.org/wiki/>, there were other facilities to foster community participation—via forums (<http://www.ros.org/wiki/>) and meta blogs (<http://planet.ros.org/>)—very much mimicking the approach of large open source projects such as Ubuntu, Python, Apache, etc.

Brian Gerkey, chief developer of ROS, explains ROS as follows: *ROS = plumbing + tools + capabilities + ecosystem.*